Assignment for Week 8

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Package Loading

```
library(ggplot2)
library(dplyr, warn.conflicts = F)
```

Assignment

1) 5.1 (p 284)
2) 5.8 (p 290)
3) 9.20 (p 518)
4) 9.22 (a) (b) (p 519)
5) 9.26 (p 519)
6) 9.27 (p 519)
7) 9.30 (p 520)
8) Suppose $\bar{X}$ is the mean of 100 observations from a popualtion with mean μ and variance σ^2. Find limits between which $\bar{X} - \mu$ will lie with a probability of at least .85 using the Central Limit Theorem.
9) Consider the function $f(x, y)$ defined by the following table

Table 1: $f(x, y)$

	y= 1	2	3
x= 1	.1	0	.1
2	0	.2	.2
3	.1	.2	.1

(a) Verify that $f(x, y)$ is a PDF.
(b) Find the marginal probability functions of X and Y.
(c) Compute E(X) and V(X).
(d) Compute the covariance between X and Y.
(e) Find $P(X < Y)$.
(f) Are X and Y independent? Justify your answer.

R assignment

10) Write your own functional equivalents to the listed values of each of the following functions. Each should take on a vector and output one of the listed values.

a) `t.test(X,Y, var.equal=TRUE)` - statistic, parameter, p.value, conf.int.

b) `var.test(X,Y)` - statistic, parameter, p.value, conf.int.

11) What Departments have the highest gender differences in salary in Chicago? What Departments have the lowest gender differences in salary?